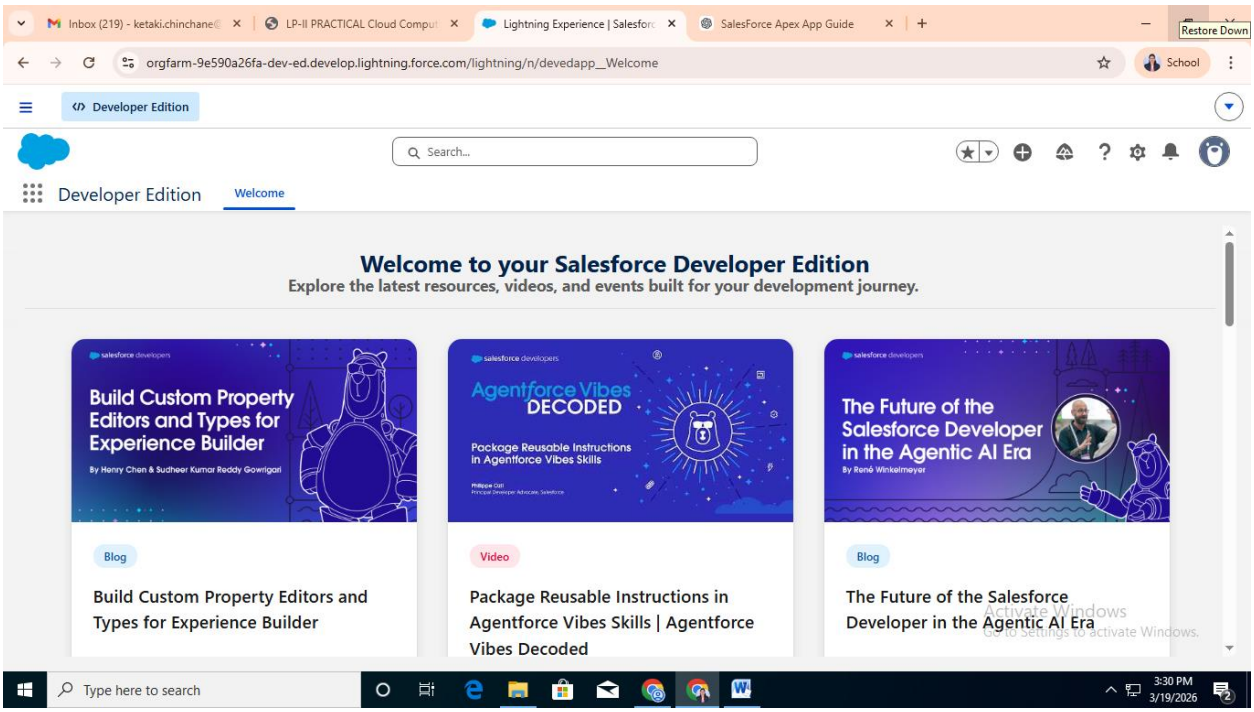
Practical 03: Creating an Application in SalesForce.com using Apex programming Language

Student Management Application

Step 1: Create a Salesforce Account and Login https://login.salesforce.com/

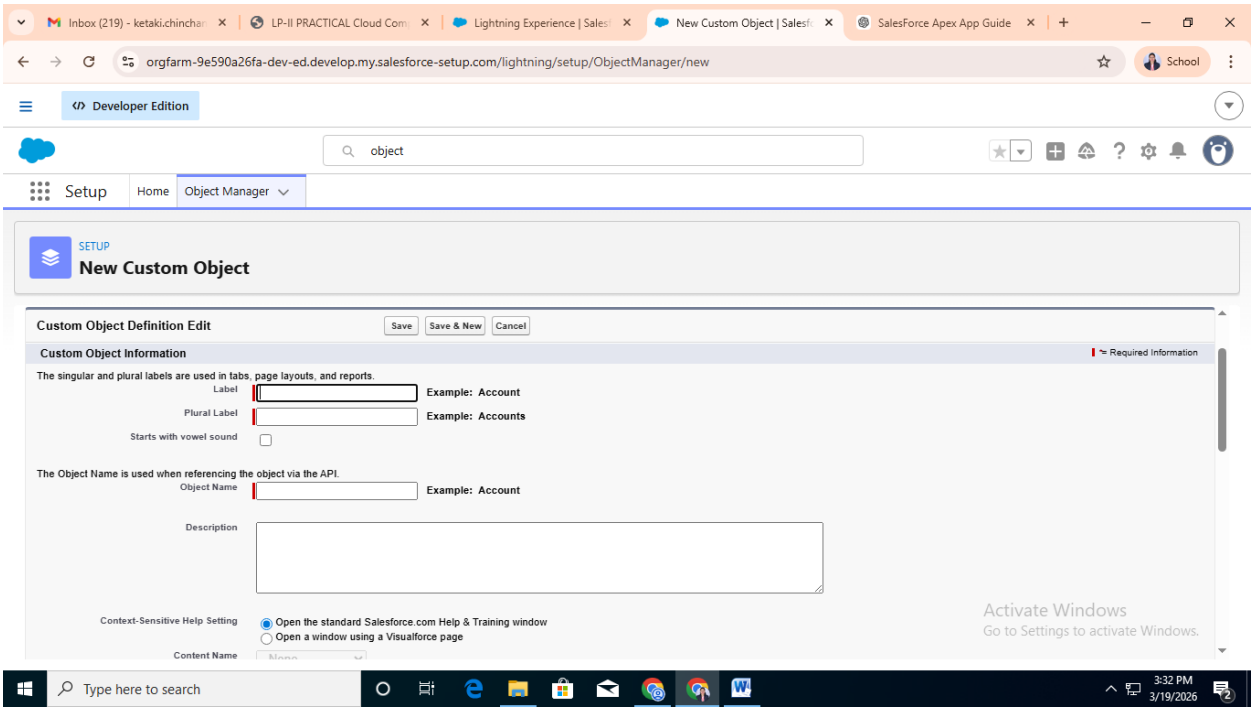


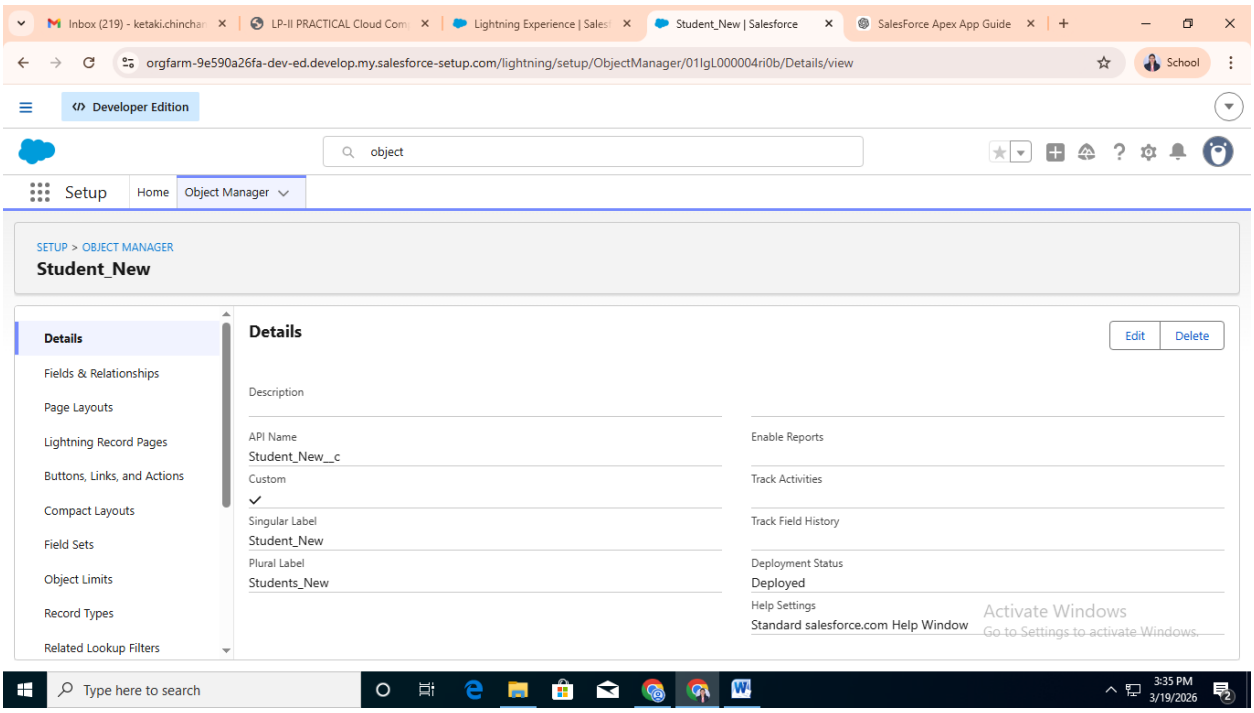
Step 2: Create Custom Object (Student)

- Go to **Setup**
- Search: **Object Manager**
- Click **Create → Custom Object**
- Fill details:

  - Label: Student
  - Object Name: Student

- Save



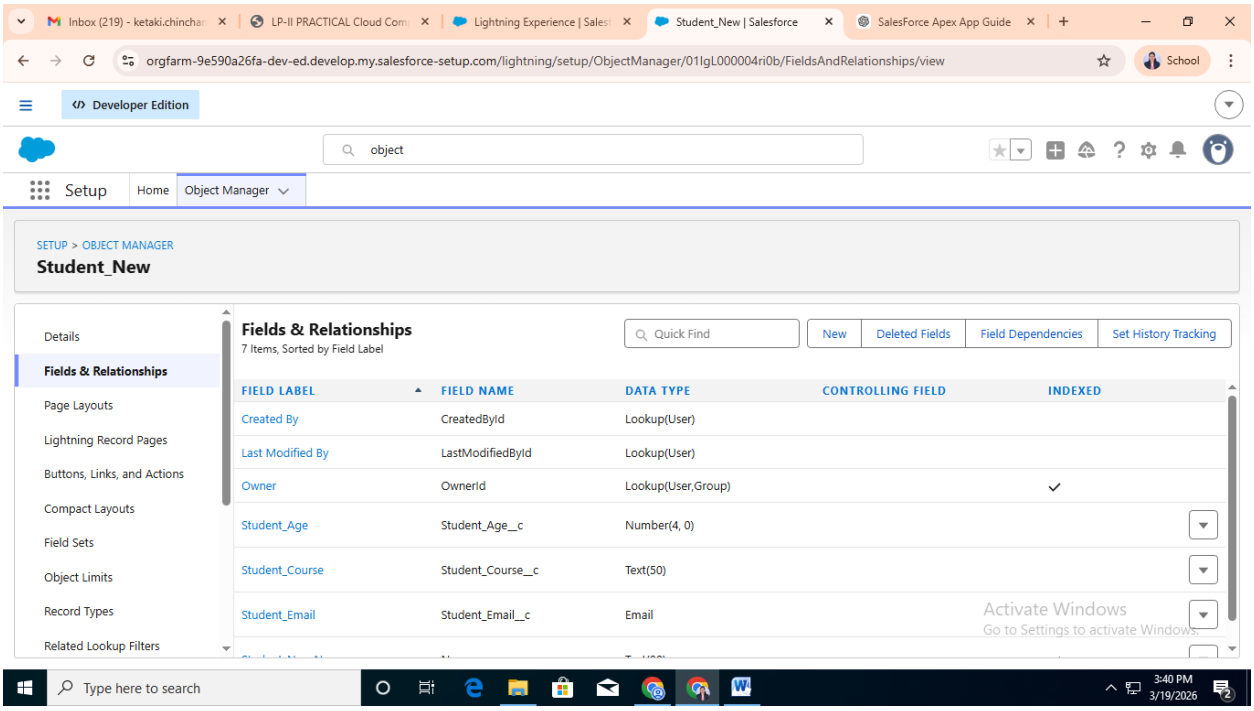


Step 3 – Add Fields:

Go to Fields & Relationship → New → Create all fields below one by one.

- Name (default)
- Age (Number)

- Email (Email)
- Course (Text)



Step 4: Create Apex Class

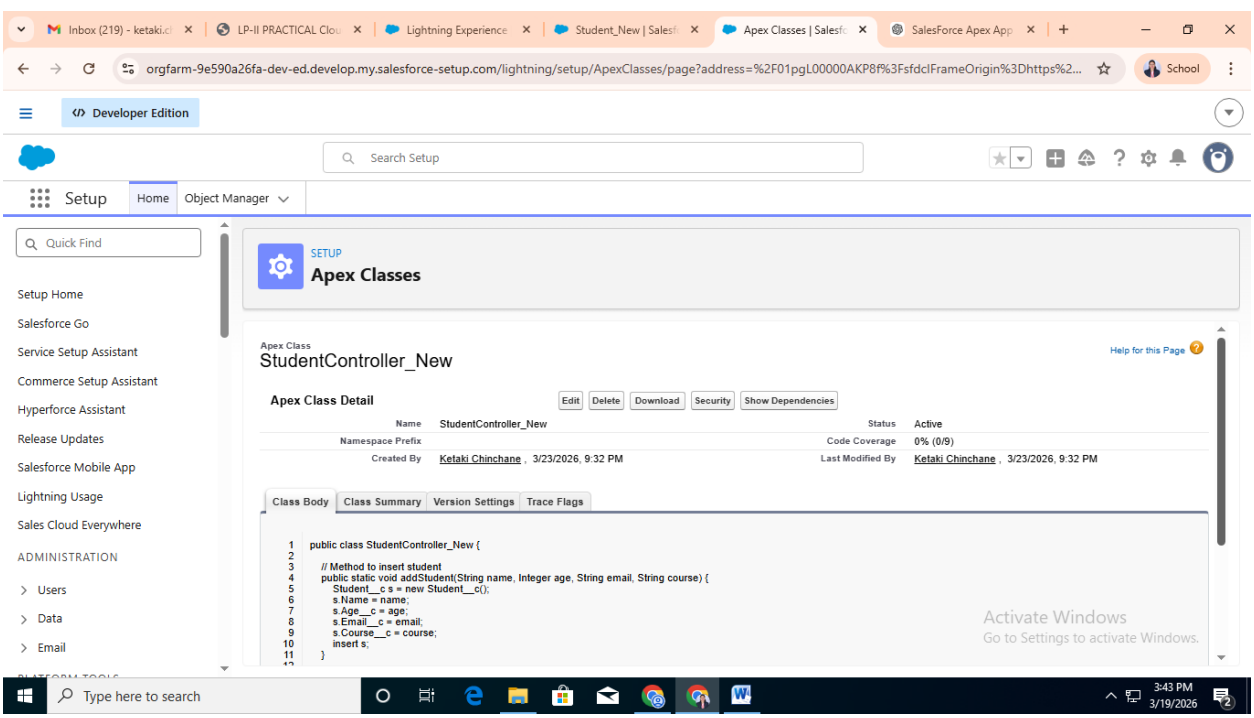
1. Go to **Setup → Apex Classes**
2. Click **New**
3. Write this code:

This class:

- Inserts student data
- Retrieves student records

```
public class StudentController {


  // Method to insert student

  public static void addStudent(String name, Integer age, String email, String course) {

    Student__c s = new Student__c();
```

```
        s.Name = name;
        s.Age__c = age;
        s.Email__c = email;
        s.Course__c = course;
        insert s;
    }

    // Method to get all students
    public static List<Student__c> getStudents() {
        return [SELECT Id, Name, Age__c, Email__c, Course__c FROM Student__c];
    }
}
```



Step 5: Add Validation Rule (Instead of Trigger)

We'll enforce the "Age ≥ 18" condition using **no-code validation**.

Steps:

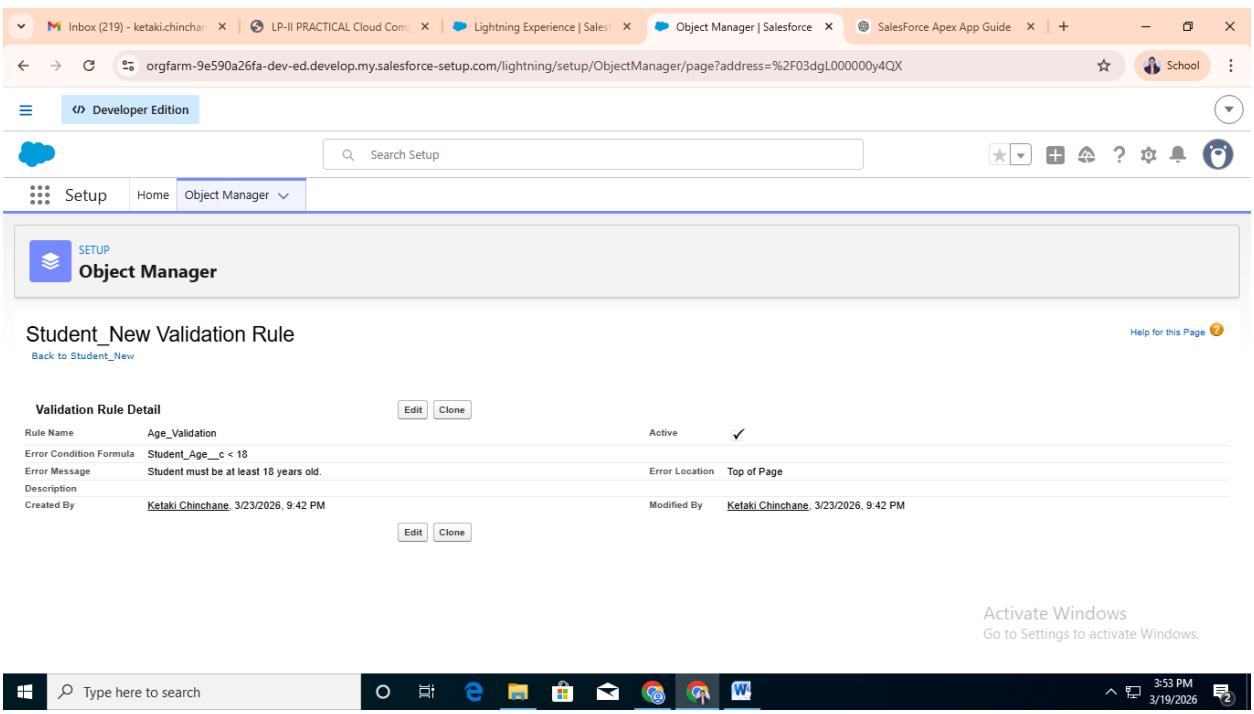
1. Go to **Setup → Object Manager → Student**
2. Click **Validation Rules**
3. Click **New**

Enter:

- **Rule Name:** `Age_Validation`
- **Formula:**

```
Age__c < 18
```

- **Error Message:**
  `Student must be at least 18 years old.`



Step 6: Test Your Apex Class

Now test your class from **Developer Console**.

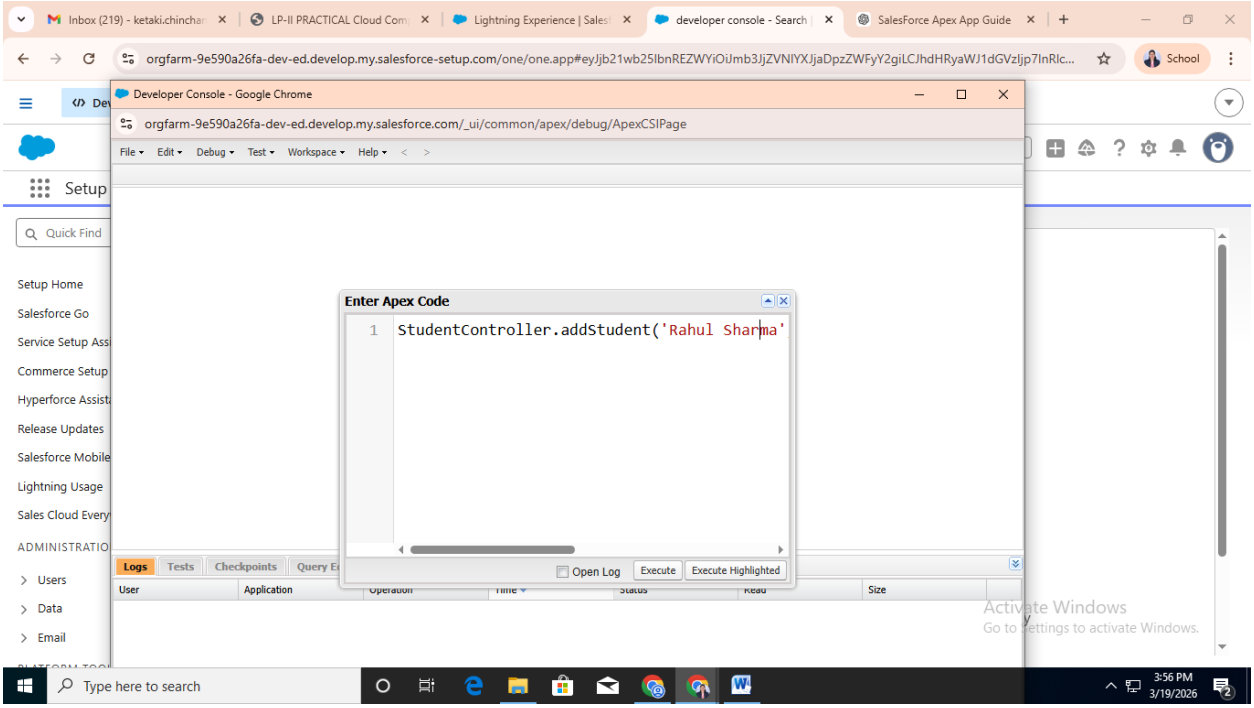
Steps:

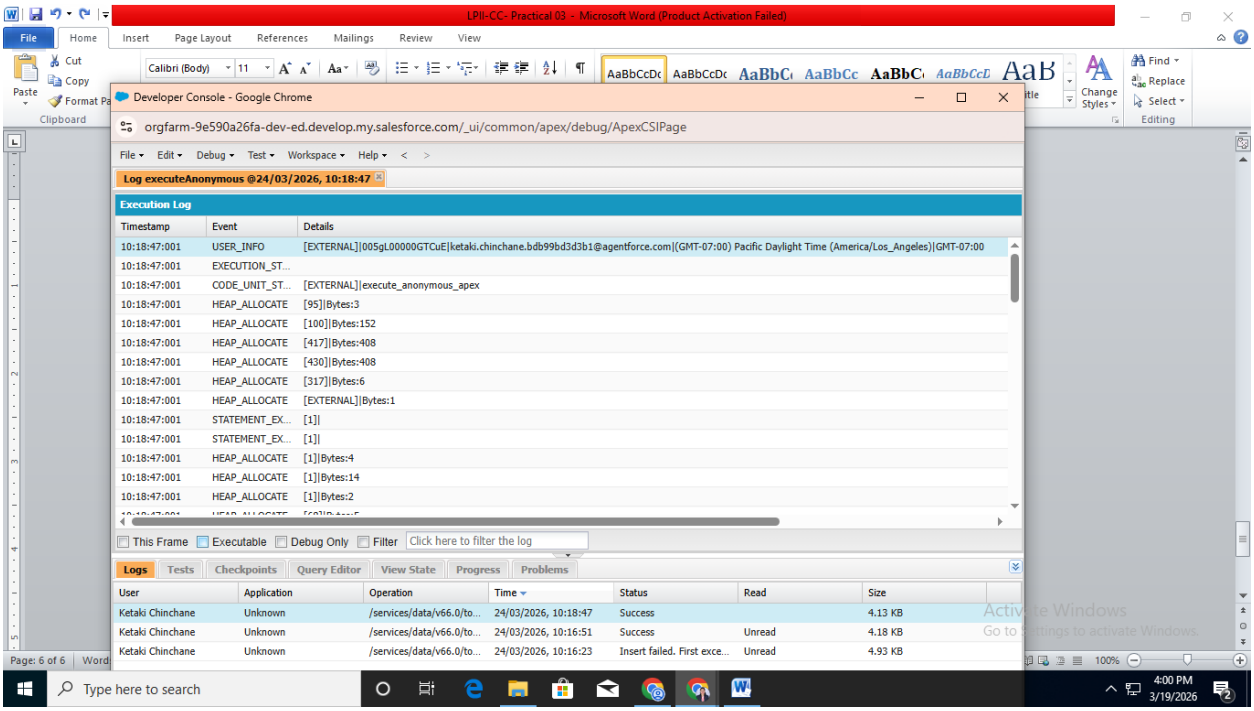
1. Go to setup button but you will find **Developer Console below setup icon.**
2. Go to **Debug → Open Execute Anonymous Window**
3. Run:
4. **Valid case:**

   StudentController.addStudent('Rahul Sharma', 20, 'rahul@email.com', 'Computer Engineering');

   Invalid case:

   StudentController.addStudent('Amit', 16, 'amit@email.com', 'IT');





Step 7: Create a Custom App (UI)

Now let's build a simple UI.

Steps:

1. Go to **Setup → App Manager**
2. Click **New Lightning App**
3. Enter:
   - App Name: `Student App`
4. Click **Next → Next**
5. Add **Student Object**
6. Finish